Addressing Data Gaps:
Four New Lines of Inquiry in the 2017 Economic Census

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Abstract

We describe four new lines of inquiry added to the 2017 Economic Census regarding (i) retail health clinics, (ii) management practices in health care services, (iii) self-service in retail and service industries, and (iv) water use in manufacturing and mining industries. These were proposed by economists from the U.S. Census Bureau's Center for Economic Studies in order to fill data gaps in current Census Bureau products concerning the U.S. economy. The new content addresses such issues as the rise in importance of health care and its complexity, the adoption of automation technologies, and the importance of measuring water, a critical input to many manufacturing and mining industries.

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1. Introduction and Background on the Economic Census

The Census Bureau continually updates and improves its measures of the dynamic U.S. economy. Researchers at the Center for Economic Studies (CES) contribute to this process by undertaking research using business microdata. This research leads to discoveries in economics and to unique insights into the quality of Census Bureau data. CES research identifies potential improvements to current data collection or survey methodology and the resulting enhancements (for example, adding new questions on surveys) lead to more relevant measures of our dynamic economy. This paper describes four additions to the recent Economic Census (EC) proposed by CES, but it should be noted that these changes represent only some of the many improvements to the EC.[1]

Collected every five years, in years ending in 2 and 7, the EC compiles statistics on approximately seven million employer business establishments from all sectors of the economy, except agriculture and public administration. The EC is part of the foundation of the nation's system of economic statistics and the primary source of information about the structure and functioning of the nation's economy. Statistics from each EC are an important part of the framework for the National Income and Product Accounts, input-output tables, and various economic indices, including gross domestic product and the producer price index. It also provides the sampling frames and benchmarks for a number of business surveys that track higher-frequency economic activity. Statistics from the EC are used by federal, state, and local governments in policymaking, planning, and public administration; by businesses, trade associations, and chambers of commerce for development and business decisions; and by academia and the general public to better understand our economy.

The EC produces basic statistics on the number of establishments, revenue, payroll, and employment, as well as details on materials, fuels, and electricity consumed, depreciable assets, selected expenses, inventories, capital expenditures, and revenue by product line. Statistics are published for over 950 detailed industries; for 7,900 goods and services; for the United States, each state, and for nearly 21,000 smaller geographic areas, including the U.S. territories; as well as by type of operation, business size, and other measures.

The EC also collects special sector- or industry-specific data, and it is new content of this type that we highlight in this paper. In particular, the 2017 EC debuted questions on retail health clinics (i.e., clinics located in pharmacies, supermarkets, general-merchandise stores, and the like), management practices in health care facilities (e.g., hospitals, doctors' offices, medical labs, nursing homes), the presence of self-checkout and other self-service options at retail establishments and restaurants, and water use and recirculation at manufacturing and mining facilities.

[1] An exhaustive list of these changes is beyond the scope of this paper. Improvements include the introduction of the North American Product Classification System, addition of collections on business cooperatives, use of cloud computing, and an expansion of factoryless goods production inquiries. For a full list of additions, deletions, and modifications, see Attachment J of the supporting materials associated with Federal Register (2017).

In each of these cases, CES economists identified a data gap through their research, an evaluation of available data, and consultation with stakeholders at other federal agencies. These efforts were part of a broader Census Bureau effort to reach out to stakeholders for input on collections. In the years before each EC, the Census Bureau solicits suggestions on questionnaire content and design from other federal agencies, advisory committees, industry groups, and others. The Census Bureau and the U.S. Office of Management and Budget (OMB) jointly evaluate data requests and approve them using several criteria. These include their necessity to important stakeholders; unavailability from other sources; collection, processing, and dissemination costs; the ability of businesses to supply the requested data; and the reporting burden placed on respondents.[2]

All newly proposed questions (and their instructions) are reviewed by survey methodologists at the Census Bureau who are experts in cognitive issues. The questions are also pretested in qualitative cognitive interviews with a small number of respondents. These activities often lead to redesigned questions, refinements in wording, and additions or clarifications to survey instructions. In certain cases, a recommendation to abandon a question may result.

Following this development process, the Census Bureau finalizes drafts of questionnaires and publishes a final *Federal Register* notice of its intent to submit the EC to OMB for review and approval. In the case of the 2017 EC, this occurred on March 10, 2017 (Federal Register 2017). The notice provides web links for the public to review proposed questionnaire additions, deletions, and changes to existing questions. The public is then given 30 days to submit written comments and recommendation to OMB regarding the proposal. Final approval of all census forms rests with OMB.

The 2017 EC utilized over 800 unique survey forms, covering over 950 detailed industries, to collect 2017 year-end data from approximately 4 million business locations. (Administrative records are used to compile statistics on the remaining businesses that are not sent survey forms.) Data collection started in February 2018 and concluded in March 2019, with data releases scheduled to begin in September 2019 and continuing through December 2021. For the first time ever, data collection was completely via the internet.[3]

In the remainder of this paper, we present in turn each of the four new lines of inquiry that CES proposed and facilitated. For each, we discuss (1) the data gap being addressed and the main stakeholders, (2) the new questions and their scope, and (3) post-collection plans. These new data collections will be subject to evaluation and validation exercises by CES economists, who will work with interested parties to evaluate the quality of the data and develop suitable edit and imputation methodologies for these items, with the goal of releasing experimental tabulations where possible. The results of these exercises will be made public through CES

[2] See U.S. Census Bureau (2011) for a discussion of the process of developing EC questionnaires.
[3] Exceptions were made for the Island Areas, including Puerto Rico and the U.S. Virgin Islands, which have more limited internet access.

working papers. In addition, we expect the EC microdata to be available to qualified researchers on approved projects through the Federal Statistical Research Data Center (FSRDC) network; the work of these researchers also yields improvements to Census Bureau products and new insights into the U.S. economy.[4]

2. Retail Health Clinics

2.1 Background and Stakeholders

Health care accounts for a large and growing component of our nation's economy, with expenditures totaling $3.2 trillion in 2016, according to the Centers for Medicare and Medicaid Services (CMS). Historically, the Census Bureau has collected and published data on providers in the *service* sector, such as physicians' offices, clinics, and hospitals. However, an alternative, relatively new delivery model has emerged in the retail trade sector: retail health clinics (RHC). An RHC is an in-store clinic with a health care professional who provides medical care such as vaccines, health screenings, treatment of minor injuries and illnesses, and management of medications and treatments. Understanding the number and location of RHCs, their operational structure, the extent of their activity, and the services they offer is important, but limited data currently exist on these relatively new health care providers, a hybrid bridging the retail and service sectors.

The first RHC opened in 2000 and since then the number of RHCs has increased over time, with an estimated 2,800 locations in 2017.[5] These clinics are typically located in pharmacies, supermarkets, and general-merchandise stores, including warehouse clubs and supercenters. RHCs treat minor acute conditions, provide screening and diagnostic services, are usually staffed by nurse practitioners or physicians' assistants, and, by definition, are not freestanding (Weinick et al. 2010).

RHCs differ from other types of clinics that have opened in recent decades. Urgent care centers (UCCs), like RHCs, offer convenient walk-in availability and extended business hours; but UCCs frequently require a board-certified physician on staff, treat acute but not life-threatening conditions (e.g., provide intravenous fluids, set fractures), and are often freestanding facilities (McNeeley 2012). Unlike RHCs, outpatient facilities, operated in connection with a hospital,

[4] To learn more about the FSRDCs, see www.census.gov/fsrdc.

[5] The first one, QuickMedx, opened in the Minneapolis-St. Paul area in May 2000 (becoming CVS's Minute Clinic in 2002). A father who could not get his son into a doctor for a strep throat test first introduced the retail clinic concept (McKinlay and Marceau 2008). Initially, RHCs opened with the support of venture capitalists; their customers paid for services out of pocket, and for this reason, clinics were located in affluent areas where customers could afford to pay for this convenience (Muroff 2009). Although only about 200 clinics existed in 2006, this number increased to over 1,000 by 2008 (Kaissi and Charland 2013). During the recession that followed, many clinics closed and fewer investors owned shares in RHCs. But growth again surged as hospital systems began to partner with or own RHCs, and early estimates suggested there would be more than 2,800 RHCs in 2017 (Kaissi and Charland 2013, Tu and Boukus 2013, RAND 2016).

offer services of licensed physicians in various medical specialties who diagnose and treat patients (e.g., provide surgical procedures). RHCs are also distinct from worksite clinics, which serve only employees. Finally, community health centers provide safety-net services to the uninsured and to underserved areas and are staffed by general practitioners and emergency physicians; they may offer services differing from those offered by RHCs or may open their own RHCs (Taylor 2004, Scott 2010).

RHCs uniquely deliver health care services in a retail environment and the new questions on the 2017 EC will help measure their prevalence and their services. How this emerging industry bridges the retail and health care services' sectors will be revealed with new data on their ownership and operational structure, which can be sole operation by retailers, sole operation by health care providers, or a partnership of the two. The location of RHCs is particularly relevant for evaluating their potential role in improving access to affordable health care services: proximity to a RHC is a strong predictor of health care use, RHCs offer alternatives to traditional care givers, and these clinics represent a potential source of care for the underserved (Ashwood et al. 2011, Wang et al. 2010, Pollack and Armstrong 2009). While many conditions can be treated in a RHC and do not require emergency care (Weinick et al. 2010), research examining the relationship between proximity to a RHC and emergency department (ED) visits has been inconclusive.[6]

Along with concerns about access, rising health care costs are a major public-policy concern, particularly as they impact low-income consumers. RHCs may provide a lower-cost alternative over traditional health care providers for minor conditions, although their impact on overall costs is unknown (Thygeson et al. 2008). Care initiated at a RHC is about 30-40% cheaper than similar care provided in a physician's office and about 80% cheaper than similar care at an ED (Ashwood et al. 2011).[7] Due to increased insurance coverage for RHC services, patients' out-of-pocket payments dropped from 100% in 2000 to 16% in 2007 (Mehrotra et al. 2008, Rudakvsky et al. 2009).[8] On the other hand, at least as of 2014, most RHCs were still located in higher-income non-rural areas with higher concentrations of white residents and fewer black and Hispanic residents (RAND 2016). Using Census household-income data for urban areas, Rudavsky and Mehrotra (2010) find that the population within a five-minute driving distance from a RHC has a higher median household income, is better educated, and is as likely to be insured as the rest of the urban population.[9]

[6] Some find that close proximity to a RHC reduces the likelihood of visiting an emergency department (ED), and others find no reduction in ED visits for low-acuity conditions (Alexander et al. 2017, Martsolf et al. 2017).
[7] Using 2005-2006 claims data, Mehrotra et al. (2009) find that RHC costs are 30% lower than for UCCs for three common acute conditions (otitis media, pharyngitis, and urinary tract infection).
[8] By 2008, most RHCs accepted some form of public or private insurance, but many RHC visits were still paid out-of-pocket (Rudavsky et al. 2009, Mehrotra and Lave 2012).
[9] Although RHCs provide care at a lower cost than other health care sites, RHCs can increase health care spending by increasing utilization (Ashwood et al. 2016). Cost savings from use of RHCs may also be limited by scope-of-practice laws for nurse practitioners, who often staff RHCs when permitted under state law (Spetz et al. 2013).

The new data on RHC locations will open up possibilities for studying their impact on community health care outcomes. In addition, this new retail information may support Census stakeholders' measurement of health care spending not currently included in CMS's National Health Expenditure Accounts and/or the Bureau of Economic Analysis's (BEA) measurement of personal consumption expenditures. Although RHC use has been low relative to other types of outpatient health care encounters, there were nearly six million RHC visits to the three largest retailers in 2009 (Rudavsky et al. 2009). This suggests that measuring personal consumption of RHC services for health care expenditures may not be trivial.

2.2 New Questions: Retail Health Clinics

Table 1 presents the RHC questions added to the 2017 EC. These questions are directed to supermarkets, pharmacies and drug stores, department stores, and warehouse clubs and supercenters. First, these retailers are asked whether they operate or lease space to a retail health clinic, and are instructed to skip all remaining questions on RHCs if they respond 'no.' For retailers that operated RHCs, Question 2 asks whether they operate their own clinic, lease space to another business or health system that operates the RHC, or partner or jointly operate the RHC with another business or health system. RHC operational models have evolved since they first opened in 2000, and these new data will provide a snapshot of their structure in 2017.

An RHC's operational model affects the retailer's degree of control of medical oversight and revenues. Retailers that operate their own RHCs retain all revenues. Retailers jointly operating a RHC with other businesses, including health care providers, may control the RHC's finances while medical oversight and follow-up care may be provided by a health care provider. Finally, some retailers may lease space to other businesses or health systems that operate the RHC. As a lessor, the retailer is likely to have limited, if any, knowledge of the RHC's revenues and services and is not asked any of the remaining questions.

Those retailers that do operate their own RHC or jointly operate a RHC with another entity are asked three additional questions about revenues, medical services provided, and technologies used in service provision. Question 3 asks total revenues from patient care services, and provides instructions on including total patient care operating receipts for medical services such as flu shots and revenues from public and private payers.

Question 4 asks RHC operators to select the medical services they provide from a checklist of options. The checklist reflects trends in RHC utilization over their 17-year history and possibly into the future. Whereas the first RHCs mostly treated mild acute conditions, such as sore throats and skin conditions, Mehrotra and Lave (2012) report that preventative care services increased by 2009. These can include immunizations such as flu shots, but also laboratory tests and biometric screenings such as blood pressure readings. In 2010, RHCs expanded their offerings to chronic disease conditions (Mehrotra and Lave 2012).[10] Here, retailers are asked whether they provide screening, monitoring, and/or management of chronic disease conditions

[10] Chronic disease conditions accounted for 86 cents of every health care dollar spent in 2010 (Gerteis et al. 2014).

such as hypertension, diabetes, or elevated cholesterol levels. To capture services related to more recent transitions in health care markets and priorities, retail operators are also asked about less common health care services, including behavioral health screenings, weight management programs, integrated pharmaceutical services, and assistance with health insurance coverage options.

Finally, Question 5 asks retailers involved in operating a RHC about the technologies they have adopted. Some RHCs operate with electronic health records that provide a patient's medical history and support collaboration between RHCs and other health care providers. Similarly, electronic prescribing (e-prescribing) supports direct transmission of pharmaceutical orders from the patient's health care provider to the retailer. RHCs can also connect patients with remote primary care physicians and specialists through telemedicine. By using telecommunications, RHCs can support remote diagnosing, monitoring, and treatment of acute and chronic medical conditions; this technology holds potential for improving access to health care in underserved areas (e.g., rural locations). Other emerging technologies such as mobile coaching applications (apps) can remotely monitor patient compliance with treatment, and patient interview software can provide clinicians with pertinent details from a patient's answers using an electronic query system.

Much of the recent growth in RHCs involves partnerships between health care providers, such as hospital systems, and retailers in operating a RHC or operating their own RHC (Kaissi et al. 2016, Convenient Care Association 2017). To help avoid potential double counting of revenues across the service and retail sectors, select ambulatory health care providers are also asked whether they were involved in the operation of a RHC. The question and the queried industries are listed at the bottom of Table 1.

2.3 Post-Collection Plans: Retail Health Clinics

This is the Census Bureau's first collection of data on RHC activity, and the quality of the data must be evaluated prior to their use in official publications.[11] In order to provide publicly available information from this particular collection, a CES working paper will be published that includes information, where possible, on the number of retail establishments operating or leasing space for a RHC, who operated the RHC, care services available, and the use of electronic health records, telemedicine, and other technologies. We will produce tabulations by geography, industry, and employment size, where possible. Additionally, the tables could include data quality information in the form of response coverage for the different RHC questions. These new data will provide increased understanding of RHCs' prevalence, location, services provided, technology used, and organizational structure.

By combining these new data on RHCs with information on population characteristics and details on the location of more traditional health care providers, as well as insurance claims

[11] Validating these new data on RHCs is limited by the lack of similar national-level data for 2017 from other sources.

data, these data can be used to evaluate RHCs' potential for improving access to affordable health care, but also the economic activity of RHCs relative to nearby traditional providers. Understanding access, utilization, and cost issues will be improved by these new data, which potentially will support expanded measurement of health care expenditures by CMS and BEA. In addition, these data will help describe the organizational structure of these relatively new retail-health care businesses. By merging these data on different RHC ownership and operational characteristics with EC measures of revenues, expenses, size, and location, more can be learned about the organization and operation of RHCs in 2017. The results from this data collection will also inform future discussions on the proper industrial classification of establishments.

3. Management Practices in Health Care Services

3.1 Background and Stakeholders

A growing literature has demonstrated the importance of structured management practices, such as performance monitoring, target setting, and incentives, on business performance, including productivity, innovation, and employment growth (e.g., Bloom and Van Reenen 2007; Bloom et al. 2013). While many studies have focused on manufacturers, the health care sector has also received significant attention. In the U.S., where health care expenditures are now nearly 18% of GDP (CMS 2019), there is a particular interest in understanding the determinants of both the cost and quality of care, and the use of structured management practices by health care providers may play a role in both.

For example, existing studies have shown that the use of more structured management practices in U.S. hospitals is associated with better financial performance, including higher productivity, profitability, sales growth, market value, and survival (Bloom et al. 2014; Bloom, Sadun, and Van Reenen 2014; World Management Survey 2010).[12] For substance abuse treatment centers, results are more mixed, with more structured practices positively associated with higher revenue per employee but unrelated to operating margins (McConnell et al. 2009).

These same studies find that management practices in health care services can vary by ownership type, business size, and market characteristics. For example, cross-country data show that publicly run hospitals have less-structured management practices than private hospitals, but for-profit and not-for-profit hospitals have similar practices. Higher management practice scores are also found in larger hospitals, consistent with economies of scale in implementing such practices, a greater need for formal practices in larger hospitals, and a causal relationship between more intensive management practices and greater demand. These

[12] Business management practices include meeting financial and operational targets, monitoring performance, and using incentives and managing human resources. The above authors use data on these practices to construct scores ranging from one to five – with one indicating no explicit, formal, or frequent use of these management practices, and five reflecting intensive use or "structured management practices."

studies also suggest that the presence of more competitors might motivate greater efficiency and promote the utilization of more modern management practices.

Certain management practices may also result in better patient outcomes. For example, more structured management practices in hospitals have been found to be associated with higher survival rates from heart attacks and general surgery, and with other productivity indicators such as shorter lengths of stay (Anderson et al. 2003, Bloom et al. 2015, Tsai et al. 2015, Chandra et al. 2016, McConnell et al. 2013).[13] In substance abuse treatment programs, more structured management practices are strongly associated with improved intake processes, which can increase the likelihood that patients will enter and remain in care (McConnell et al. 2009).

To help fill a void in our understanding of management practices by individual providers of ambulatory, hospital, and residential health care, new content for the 2017 EC was adapted from the Census Bureau's Management and Organizational Practices Survey (MOPS) of manufacturing plants (see Buffington et al. 2017, U.S. Census Bureau 2018). In particular, the 2015 MOPS asked respondents where production display boards and key performance indicators are located at the establishment, who is aware of production targets, who selects the targets, and how frequently they are reviewed. These questions are also related to performance tracking and review questions asked of hospitals in the World Management Survey (WMS), which measures who reviews performance indicators and how frequently.

Some data related to management practices in the health care sector are being collected by other federal agencies. Surveys by the National Center for Health Statistics (NCHS) collect data on staffing, patient safety, and the use of health care technologies in ambulatory, hospital, and residential health care settings. The Agency for Healthcare Research and Quality's Medical Office Survey on Patient Safety asks medical providers about their use of clinical data for quality improvement, and its Hospital Survey on Patient Safety includes questions on how managers consider suggestions, communicate feedback on errors, and prioritize patient safety. In addition, the American Hospital Association's annual survey asks about commercial contracts with payments linked to quality, safety metrics, and whether the hospital conducts internal surveys on quality and safety performance.[14]

The new data collection will complement and expand on these collections by providing more and better data on management practices that use larger samples, cover more health care industries, and can be combined with other information on these facilities. Thus these data will help further our understanding of the effects of management practices on an organization's financial performance and the quality of patient care offered. It is with this backdrop that the three questions discussed below were added to the 2017 EC.

[13] In addition, data on U.S. cardiac care units show that higher management scores and better performance on publicly reported quality measures are positively associated with patients' preference for a hospital (McConnell et al. 2016).

[14] Other agencies also collect data on related management concepts including quality improvement and performance, and some of this information complements the new questions in the 2017 EC.

3.2 New Questions: Management Practices in Health Care Services

Table 2 presents the three questions on management practices added to the 2017 EC, which focus in particular on the selection, tracking, and review of clinical performance measures. Table 2 also shows that these questions are directed at all types of ambulatory care providers, hospitals, and residential care facilities. Collectively, these industries accounted for more than 60% of the $3.2 trillion spent on health care in 2016.

In Question 1, health care service providers are asked who at the organization sees measures of clinical performance. Here, measures of clinical performance are defined as counts, incidence rates, and other measures of specific clinical processes and outcomes, which can differ across providers. For example, a key performance indicator for physicians may be the rate of patients accessing services in an emergency department for conditions that would generally be treatable in an office setting. Home health care providers may monitor the rate of unplanned acute care hospitalizations, and ambulance services may examine response times to emergencies. Hospitals may focus on medication errors or survival rates for heart attacks, while residential care facilities, such as nursing homes, may monitor the percent of long-term residents with pressure sores or those experiencing falls resulting in a major injury. For this question, response options include managers, employees (non-managers), patients and their responsible parties, and "on public display." Respondents may select more than one of these.

Question 2 asks health care providers who chooses which measures of clinical performance to collect. Response options are managers at this establishment, managers at other establishments and/or headquarters, insurance providers, government regulators or agencies, and the board of directors. Again, respondents may select more than one of these.

Finally, Question 3 asks respondents how frequently senior management reviews the measures of clinical performance: yearly or quarterly, monthly or weekly, daily or more often, or never.

3.3 Post-Collection Plans: Management Practices in Health Care Services

CES economists will evaluate the quality of these new data, including response rates and validation results using prior collections on management practices. For example, the new 2017 EC data on how frequently clinical performance measures are reviewed can be validated using similar content asked in the 2015 Annual Survey of Entrepreneurs covering all industrial sectors.

A CES working paper will present tabulations and discuss findings on management practices in health care services as feasible based on data quality. For example, like MOPS, management practice scores with standard errors, by industry, by state, and by establishment size may be constructed and disseminated conditional on data quality.[15]

[15] See www2.census.gov/programs-surveys/mops/tables/2015/mops-survey-tables/mops_survey_tables.pdf.

In addition to plans for making these data publicly accessible through tabulations and a working paper, the microdata can support empirical analyses. CES and FSRDC researchers using these new data on management practices, in combination with other internal as well as external data, could study relationships between management practices and financial and clinical performance. Using other Census Bureau data, researchers could examine how management practices relate to organizational structure, including ownership (e.g., public vs. private, tax exempt vs. not), employment size, and productivity (e.g., revenues per employee). Data on business location would support market-level analyses for ambulatory providers, hospitals, and residential care facilities.

In addition, NCHS data on these same respondents would provide additional information on patient and provider characteristics, utilization, diagnoses/procedures/treatments, and technology adoption (e.g., electronic medical records). These new data on the 2017 EC could also be combined with CMS's publicly available databases Physician Compare, Hospital Compare, and Nursing Home Compare to evaluate relationships between management practices and clinical performance. Along with geographic location, these datasets contain detailed information on various measures of quality performance for physicians, 30-day mortality measures and patient experiences for hospitals, and health inspection and clinical data for nursing homes.

4. Self-service and Extra Services in Retail and Service Industries

4.1 Background and Stakeholders

4.1.1 Self-service

In order to understand production processes and measure productivity, it is crucial to correctly measure inputs into production – labor, capital, materials, energy, and services. In a period of technological change, capturing changes in production processes can also be crucial for understanding changes in labor demand. For example, in the past several decades, many retail- and service-industry innovations have replaced *paid* labor (employees) with some combination of technology and *customer* labor. In the retail sector, the most prominent examples are self-service gas stations and self-checkout at supermarkets and drug stores. In addition, restaurants are increasingly introducing self-ordering and self-payment technologies. Examples in other sectors include automated teller machines (ATMs) by banks, self check-in by airlines, and self-checkout at libraries.[16]

Although much has been written about the impact of automation on employment (see, e.g., Autor 2015), particularly in the manufacturing sector, very little is known about the prevalence and impact of self-service, or customer-employee substitution, in retail trade and services. The

[16] Bessen (2015) examines the adoption of ATMs.

impact of these innovations is potentially quite large, as employment in retail trade and services accounts for a large share of total U.S. employment.

Self-service has been around for many years, in various settings, and in some cases, the Census Bureau has attempted to measure its adoption and diffusion. We briefly describe two such efforts before turning back to current issues. In 1916, Piggly Wiggly introduced the first self-service grocery store. Prior to this innovation, clerks behind counters served customers the goods that they requested. The first Census of Distribution, conducted in 1929, included a yes/no question on self-service on both the grocery store form and the general long form.[17] More recently, the Census of Retail Trade (CRT) measured the introduction of self-service pumps at gas stations from 1972 to 1992. Over this period, the share of gas stations providing some self-service sales of gasoline increased from 8% to 80%. Basker, Foster, and Klimek (2017) use CRT microdata from 1977 to 1992 to estimate the extent to which customer labor substituted for paid labor in the production of retail gasoline and find that approximately one quarter of the work previously done by station attendants shifted to customers when stations converted from full- to self-service pumps. This conversion was associated with a price decrease of approximately 5%, compensating customers for their effort.[18]

One of the current forms of self-service adoption that we now measure is the rise of self-checkout in retail trade. The introduction of self-checkout in grocery stores dates to 1992 (Fountain and Goldstein 2016).[19] Kroger, one of the nation's largest grocery chains, installed its first self-checkout stand in 1995 (Zimmerman 1995) in the hopes that these stands would "help […] cut costs and help customers check groceries more efficiently" (Peale 1999). A challenge is that the technology for self-checkout is continually evolving. In the U.S., self-checkout generally refers to a separate lane or lanes at the checkout, at which a scanner faces the customer, who scans items from her shopping cart before bagging them. The customer then uses the same apparatus to pay for her purchases, with cash or a credit or debit card. However, some stores have installed "scan as you go" self-checkout in which customers use a store-provided handheld device to scan items as they retrieve them from shelves, scan a final barcode to complete the transaction, and pay at self-checkout registers (Kellett 2009). Other systems forgo the checkout entirely and allow customers to pay with their cellphones (Zumbach 2016). More innovative checkout technologies are currently being developed. Amazon.com has introduced a checkout system in its "Amazon Go" stores that uses an app to automatically detect what items the customer puts in her cart – and which she returns to the shelf – then charges the customer automatically when she leaves the store (Leswing 2017).

[17] Unfortunately, the answers to this question were not tabulated in any official publications. Basker, Vickers, and Ziebarth (2018) digitized some records from the National Archives from the 1929 Census of Distribution. In the digitized sample, very few stores responded in the affirmative to the self-service question.

[18] In 1992, the CRT also included a question on self-service by shoe stores. Responses were never tabulated and the question was not repeated in later years.

[19] Hughes (1989) reports on an Ahold store in Holland that experimented with self scanning even before the introduction of the technology in the United States. Vending machines, the earliest form of self-service, have been around since the 1890s (Basker 2016).

At present, there is no reliable estimate of the extent of adoption of self-checkout technology, nor whether it is still expanding. Several large retailers have eliminated or curtailed their self-checkout stands, even as others continue installing them. For example, Lutz (2013) reports that Costco is "eliminating self-service checkout from stores because […] employees do a better job."

As the retail sector transitions from checkout systems that use employees to those that use customers, the total labor (employee plus customer) input may change. Total labor input may *decrease* if the use of customers reduces idle time: an employee may have some periods during which no customers show up, whereas the customer-worker is only there when she needs the service.[20] Conversely, total labor input could *increase* if the customer is less efficient than a cashier; an experienced cashier can scan items and handle mishaps much more quickly than an inexperienced customer.[21]

If the total labor input in the two settings is similar, the difference between the number of employees needed to process a certain number of transactions in an employee-staffed checkout and the number of employees needed to process the same number of transactions in a customer-staffed checkout provides a first approximation of the labor input from customers. As the retail sector transitions towards more-automated systems such as the "Amazon Go" system, total labor input will decrease. A full accounting of whose labor is being displaced requires first understanding how the production function works.

Restaurants, too, are increasingly embracing customer-employee substitution, particularly for orders and payments in limited-service restaurants.[22] One news article (Johnson 2016) names McDonald's, Wendy's, Hardee's, and Panera Bread as leaders in this new direction. All have installed, or are in the process of installing, self-ordering kiosks that allow customers to select their choice using touch-screen computers. Other restaurants have installed tablets or other devices that allow customers to pay at the table without a need for employee intermediation (Alterio 2008). Although chain restaurants' adoption of these technologies generate more headlines, single-unit restaurants have also adopted them; Alterio (2008) describes adoption of self-pay by Southeast Grille House in Brewster, New York.

The potential impact of these technologies goes beyond labor substitution. In particular, self-pay, like pay-at-the-pump before it, may encourage more consumers to switch from cash to electronic payment options.[23] Some also tout the possible security benefits: Alterio (2008)

[20] Mas and Moretti (2009), using high-frequency, cashier-level data from a large supermarket chain, show that cashiers are idle approximately 1/3 of the time on average, between 7AM and 8PM.
[21] Self-checkout is also associated with increased theft (see, e.g., Taylor 2016).
[22]The latest incarnation of this trend is relatively new, but cafeterias, fast-food restaurants, and even the "automats" of the early 1900s were early forms of labor saving in the restaurant industry (Bromell 2000). Automats were restaurants where customers retrieved their meals from windowed cubbies. The first automat in the U.S. opened in 1902, and the last closed in 1941 (Bromell 2000).
[23] Pay-at-the-pump was introduced in the early 1980s (Basker, Foster, and Klimek 2017) and has since become ubiquitous. The Census Bureau did not collect data on this innovation.

notes that paying at the table allows customers to keep credit cards in their possession, which may reduce the possibility of card duplication and identity theft. Finally, to the extent that miscommunication about orders is reduced by self-ordering at restaurants, it may affect the level of food wastage.

Ultimately, we hope that better understanding the correlation between employment and these structural changes in the retail sector will help the Census Bureau more accurately estimate employment, particularly when data are missing or deemed inaccurate for any reason. Improving these measures will also help us better understand labor market and productivity dynamics.

4.1.2 Extra services

We now turn from a discussion about reducing the amount of services available to customers to the addition of services available to customers. Returning to the gas station case, Basker, Foster, and Klimek (2017) find a positive correlation between gas stations' conversion to self-service and the introduction and expansion of co-located convenience stores. The added services offered by convenience stores require additional employees at the same time that self-service pumps reduce gas station employment.

In other examples, pre-ordering and delivery services are two margins on which supermarkets, particularly high-end supermarkets, differentiate themselves.[24] These services also require additional labor input, and again, the additional labor input is paid employees rather than customers.

As we discuss below, one industry in retail trade (grocery stores) will get both the self-service and extra services questions, which will enable us to understand some of their trade-offs. In particular, cross-tabulating extra services and self-checkout will allow us to determine whether the phenomenon of introducing additional services is correlated with self-service outside the gas station context.

4.2 New Questions: Self-Service and Extra Services

Table 3 presents the self-service and extra services questions added to the 2017 EC. One question is directed to home centers, supermarkets, convenience stores, health- and personal-care stores (including pharmacies and drug stores), department stores, and general merchandise stores, including warehouse clubs and supercenters. It asks: "As of December 31, 2017, did this establishment offer a dedicated self-checkout lane for customers?" Response choices are yes and no.

[24] Similarly, deli services and selling prepared foods also require additional labor input, and also allow stores to differentiate themselves. The Census Bureau already captures the sale of deli foods.

Another question is directed to restaurants, including limited-service restaurants, and similar establishments. It asks: "As of December 31, 2017, did this establishment use electronic devices for self-service table orders and/or payment (for example, through the use of a restaurant provided electronic device at the table)?" Response choices are yes and no.

The question about extra services in the supermarket industry asks: "As of December 31, 2017, did this establishment offer preordering or delivery services by website, app, fax, phone, or other means?" Response choices are yes and no.

4.3 Post-Collection Plans: Self-Service and Extra Services

The post-collection plans for these data include research into the quality of the data and determining whether experimental tabulations are feasible. Depending on data-quality and disclosure concerns, variables that could be tabulated and published include: (i) share of establishments and employment, by industry (e.g., supermarkets, pharmacies, home centers, restaurants) that have self-checkout; (ii) share of supermarkets and their employment offering delivery and/or pre-ordering; and (iii) a cross-tabulation of the two (for supermarkets). Tabulations by geography may also be possible.

Research projects on this topic, using the confidential microdata, may estimate the degree of customer-employee substitution that can be attributed to self-service options, as well as the degree of employee-customer substitution attributed to delivery services. This research may be conducted by CES economists or by researchers using the FSRDC system.

5. Water Use in Manufacturing and Mining Industries

5.1 Background and Stakeholders

With recurring droughts and concern that precipitation patterns may be shifting away from historical norms, understanding water use and conservation efforts can help ensure that supplies meet the demands of a growing population and economy. Most public attention is paid to agricultural and residential water use; however, there are some significant water-using sectors beyond these, including manufacturing and mining.[25] However, data on who and where these large water-using businesses are; the characteristics and patterns of their water intake, (re-)use, and discharge; and their water-related costs and investments are limited or outdated.

For decades, the Census Bureau collected such data. Specifically, all but one Census of Manufactures (CM) from 1954-1987 asked every manufacturing establishment its range of water intake and occasionally related questions (e.g., on recirculation, principal source, kind of

[25] According to Maupin et al. (2014), 4.5% of water withdrawn in the United States is by the manufacturing sector and another 1.5% is by mining. (These estimates only include self-supplied water—not water purchased from utilities.) As a point of reference, residential/domestic use is approximately 7.8% of water withdrawals.

water). Those establishments reporting at least 20 million gallons of annual water intake received the Survey of Water Use in Manufacturing (SWUM) typically the following year (i.e., 1954, 1959, 1964, 1968, 1973, 1978, and 1983; the survey was discontinued for funding reasons just prior to the planned 1988 survey). The aggregate statistics from the SWUM surveys were published in the respective CM volumes. The situation was similar for mining facilities, the Census of Mineral Industries, and the Survey of Water Use in the Mineral Industries. Introducing this collection of industrial water use data, U.S. Bureau of the Census (1952, p. 3) stated: "Despite the sizable expenditures of the Federal Government in planning and developing our water resources, there is a very serious gap in the availability of accurate and reliable statistics on the use of water by manufacturing plants…. The solution to such national problems as the relocation of industry, and the pollution and shortage of water, will depend in part on the development of an accurate body of information on the industrial use of water."

Until its discontinuation, the SWUM produced statistics by detailed industry (four-digit SIC), by state, and by water use region by major industry group (two-digit SIC). The SWUM collected water intake (in millions of gallons) from five sources (public water system; surface water; groundwater; tidewater; other) by three types (fresh; brackish; salt). Respondents were also asked to report their water intake for seven purposes (e.g., process; steam electric power generation; other cooling & condensing) and by type. In addition to water intake, respondents were asked to report the gallons of gross water use by the seven purposes, where gross water use is the water intake for a particular purpose *plus* the water recirculated for that purpose. Respondents were sometimes asked about the treatment of intake and recirculated water, by method. The SWUM also collected data on water discharged into seven different points (public utility sewer; streams and rivers; lakes and ponds; bays and estuaries; ocean; ground; transfer) and by whether the water was treated or not. Respondents were also asked to report their water discharge by the water's final purpose and by whether the water was treated or not. Respondents were further asked to report their water discharge by eleven types of treatment. The 1973 and 1978 surveys also collected information on water-related expenditures — the only two years of the survey to do so — including details on water-treatment assets, investments, and operating costs. The information collected on the SWUM varied somewhat over time, in both detail and in scope. See Becker (2016) for further discussion.

What we know about current water use in manufacturing and mining does not come from such systematic data collection.[26] In its reporting guidelines to states, relating to its quinquennial report on water use in the United States (e.g., Maupin et al. 2014), the U.S. Geological Survey (USGS) encourages states to contact various state and local agencies and other entities for water use information they may hold, to purchase lists of manufacturing establishments, to conduct surveys or site visits, and/or to impute water use. In particular, the USGS guidelines (Hutson 2007, p. 14) state:

> *There are three general approaches used to compile withdrawal data by industries:*

[26] This is true for water use more generally (Fishman 2016).

1. *Acquire site information and withdrawal data for individual industries, focusing on larger ones while striving for an adequate representation of the withdrawals in each county.*
2. *Acquire site information with ancillary data on employment or production, and estimate water withdrawals using water-use coefficients. These coefficients are usually in the form of usage in gallons per day per employee or per unit of product.*
3. *Combine the two approaches by acquiring industrial facility information and water withdrawals for the larger industries and using these data to develop water-use coefficients for estimating withdrawals for the smaller industries.*

The guidelines do not specify which method dominates, but in his letter of support for the new 2017 content, the USGS Associate Director for Water, William H. Werkheiser, wrote: "Estimation of water-use rates for industrial facilities and mines is particularly challenging in states where water withdrawals are not regulated or reported. In many of these cases, the USGS estimates withdrawals based on industry or mineral-specific water-use coefficients and annual production data, some of which may be incomplete or outdated."[27] Indeed, discussions with USGS hydrologists revealed that the water-use coefficients that are used are, in part, based on water use rates implied by the Census Bureau's last SWUM (from 1983) and that "for both mining and manufacturing, there has been very little independent research or surveys that have produced usable coefficients. Regulatory agencies in some states compile water withdrawal data for mines and industries, but in other states USGS personnel must collect the data independently or estimate withdrawals, often using withdrawal rates gleaned from the old [Census Bureau] reports and minerals or manufacturing production data from more recent [Census Bureau] reports."[28] Even without these data limitations, "the use of coefficients to estimate industrial water use is imprecise because of the variability in factors affecting water use by industries. The specific processes, age of the facility, cost of the water and wastewater treatment, and amount of recycling all contribute to the amounts of water needed by an industry. These factors should be considered when using national coefficients or when developing and adjusting local coefficients for industries not surveyed" (Hutson 2007, p. 15).

However estimated, every five years, states must forward to the USGS the following five estimates of industrial water use, by county, which is stored in the USGS Aggregate Water Use Data System (AWUDS): ground-water withdrawals (freshwater, saline water), surface-water withdrawals (freshwater, saline water), and reclaimed waste water. These statistics are for all of manufacturing and lack any industrial detail.

In supporting the Census Bureau's efforts to collect water use data in manufacturing and mining, William Werkheiser of USGS mentions not only the potential to construct new water-use coefficients, but also the ability to better understand the sources of industrial water and the efficiency of industrial water use (e.g., through recirculation and reuse). In his letter of

[27] Letter from William H. Werkheiser to Lucia Foster, December 10, 2015.
[28] Personal communication from USGS National Water Use Leadership Team staff to Randy Becker, September 17, 2015.

support, Joel Beauvais, the Deputy Assistant Administrator in the Office of Water of the U.S. Environmental Protection Agency (EPA), also cites the benefit of updated water-use coefficients and adds: "Increased pressures on water supply from climate change and drought have led to water recirculation and reuse becoming more and more important as coping mechanisms for manufacturing and mining establishments to achieve a more reliable water supply. Updating the establishment-level water intake and recirculation data would ensure that future use estimates by sector would be more representative of current practices."[29] He also names specific EPA programs that can benefit from these data.

In addition to their usefulness to the work of various federal, state, and local government agencies, these data will also contribute to research. Industrial water use has received relatively little scholarly attention, relative to agricultural, residential, and recreational uses, perhaps in part because of the lack of good, comprehensive data. These new data will permit exploration of industrial water demand and the determinants of water efficiency. In one such study, using SWUM data from the 1970s, Becker (2016) finds water use was concentrated in particular manufacturing industries and that water use per unit of output was greatest in larger facilities. Larger facilities used water for more purposes but also recirculated water more – satisfying water demand without necessarily increasing water intake. (When last measured in 1983, each gallon of water taken in by the manufacturing sector was used 3.4 times on average.) Becker (2016) also finds that water recirculation was greatest when water was purchased from a utility, as opposed to self-supplied (which, at the time, was the vast majority of water used in manufacturing). Recirculation was particularly low at manufacturing plants in which self-supplied tidewater or surface water were the primary source, less so for groundwater. Relatively low costs for self-supplied water may have suppressed the incentive to invest in recirculation. Meanwhile, manufacturing plants with higher per-gallon water intake treatment costs recirculated more, as might be expected, as did those with more treatment of discharged water.

These findings may be outdated, given the shifts in the composition of U.S. industrial activity over the past four decades, changes in the geographic distribution of activity, and possible changes in how, and how much, water is used in such facilities. Though much more limited in scope than the bygone Survey of Water Use in Manufacturing, the water-related inquiries in the 2017 EC have the potential to significantly increase our understanding of who and where are (large) water-using businesses, their main sources of water, and the factors underlying their relative water efficiency.

5.2 New Questions: Industrial Water Use

While the need for better industrial water use data is clear, the resumption of a detailed survey like the SWUM is not possible at this point. Instead, we set our sights on (re)introducing a few key questions, using the 2017 EC as the vehicle. The wording and structure of these five questions, which are presented in Table 4, were developed after careful study of the Census

[29] Letter from Joel Beauvais to Lucia Foster, January 29, 2016.

Bureau's previous water use inquiries (from 1954-1983), consideration of Statistics Canada's current biennial Industrial Water Survey, close consultation with hydrologists from the USGS's National Water Use Leadership Team, and feedback from Census Bureau experts on instrument development, including cognitive and usability testing.

The first question here is the most fundamental: gallons of water intake. Historically, *water intake* was never explicitly defined on either the CM or SWUM, perhaps because it is self-evident. Here, we incorporate some clarifying language found in the Canadian survey: "new water introduced into the establishment for the first time, regardless of source or quality." When previously asked on the CM, the water intake question would always note that water used in "the production process and auxiliary operations (such as cooling and condensing, sanitary services, and boiler feed)" was to be included. Here, we use the same wording in the instructions (which in the electronic survey instrument appears on-screen at the same time as the question). Because this question will be answered by establishments of all sizes (in contrast to the SWUM, which was targeted to establishments with a minimum of 20 million gallons of water intake), the reporting unit has been lowered from millions to thousands of gallons. The maximum allowable response here is at least one order of magnitude above the expected extreme case.

Question 2 (does this establishment recirculate or reuse any water) serves as a screening question for Question 3 (gross water use in gallons). While technically unnecessary (an establishment that does not recirculate or reuse water simply has gross water use equal to water intake in Question 1), because establishments of all sizes will face these questions, and because smaller establishments (prevalent in most industries) are much less likely to recirculate water (Becker 2016), splitting the question in two in this manner has merits from a cognitive and/or response burden standpoint. It is also consistent with the manner in which gross water use (recirculated and reused water) was historically collected on the SWUM.

Though it may seem a more modern phenomenon, gross water use, which is the sum of water intake plus water recirculated/reused, was collected on the very first SWUM in 1954.[30] The definition that is embedded in the question ("the quantity of water that would have been required if no water had been recirculated or reused") is identical to one that has appeared in SWUM instructions (e.g., in 1978). As discussed in Becker (2016), the questions on gross water used and water recirculated/reused took different forms over the years. Here, there was some discussion of whether to ask for 'gross water use' or for 'water recirculated or reused' before deciding on the former.[31] The hypothetical example given in the instructions is essentially identical to the one used since the 1968 SWUM, and similar to those used prior to that. As with Question 1, the reporting unit is thousands of gallons, and the maximum response can be in the trillions of gallons.

[30] At that time, each gallon of water taken in by the manufacturing sector was used 1.82 times, on average. This is the ratio of gross water use to water intake.

[31] The Canadian Industrial Water Survey asks for 'water recirculated or reused' rather than 'gross water use', but explicitly excludes water recirculated in closed-loop systems.

Question 4 asks about the source of an establishment's water. Historically, the SWUM had asked for *quantities* from different water sources. Here, for simplicity in reporting, and recognizing that most establishments get all or most of their water from only one source, we ask only for the *main* source of water. The six categories here all previously appeared on the SWUM; there are some slight modifications, however. First, "company" water system was changed to "self-supplied" water system.[32] This is the terminology used in the Canadian survey, and it seems clearer. Likewise, the use of "own system of pumps, pipes, hoses, etc." in the definition of self-supplied water systems is new and also taken from the Canadian survey. Instructions regarding a "joint water supply system" has been modified to read "another company that is not primarily a water utility." For simplicity in programming the electronic survey instrument, the choice of "mine water" is offered to manufacturing establishments in addition to mining facilities. The examples of mine water, previously found in the instructions, were moved into the answer option itself. Meanwhile, the SWUM never defined "other sources." Here, a couple of examples from the Canadian survey are offered. The definition of "public water system" now uses the word "utility" (as does the Canadian survey) rather than "system", but is otherwise similar to the definition on the SWUM, though much more concise.

Finally, Question 5 is one posed only to those establishments that designate *public water system* as their main source of new water. This question asks whether that water is mainly potable or reclaimed (i.e., treated wastewater that can be used for certain purposes).[33] While the SWUM asked about certain types of water from its inception – namely, fresh, brackish, and salt – it never asked about reclaimed water. This was added at the request of USGS.

At this time, there are no questions on water intake and gross use *by purpose*, water discharge, water treatment, or water-related expenditures.

5.3 Industry Selection: Industrial Water Use

Historically, water intake (in three to six different ranges) was asked of all recipients of the Census of Manufactures [Mineral Industries], regardless of detailed industry. Respondents with at least 20 million gallons of annual water intake received the follow-up SWUM, again regardless of detailed industry.

In contrast, the 2017 EC only asks these water use questions of establishments in significant water-using manufacturing and mining industries – not *all* manufacturing and mining industries. Unlike the SWUM, however, responses are obtained from water users of all sizes in these targeted industries.[34]

[32] In 1983, about 87% of the water taken in by the manufacturing sector was self-supplied.

[33] This is different from the reused/recirculated water from Question 3 in that reclaimed water is "new water" introduced into the establishment from an outside source.

[34] While water intake is especially concentration in certain industries (Becker 2016), the SWUM microdata of the 1970s indicate that 380 (84%) of the 450 manufacturing industries had at least one "large" water-using establishment (with more than 20 million gallons of water intake), as did 37 of the 42 mineral industries. At the

The main challenge here was identifying significant water-using industries in the absence of good, comprehensive, timely data on the subject. After discussion with USGS, we agreed to use the 1983 data on industry-level water intake as the foundation for selecting industries, fully realizing that industries may have increased or decreased in (relative) importance over the past three decades and that water use within any industry may have also changed in that time. The USGS would then provide further expertise to add industries and suggest prioritization, as necessary.[35]

In the end, all of the six-digit 2012 NAICS industries with at least 45 billion gallons of water intake in 1983 were recommended for data collection in the 2017 EC. Of the industries in the 17-45 billion gallon range, about one third were set aside because they were deemed a lower priority by USGS. A number of these industries have also decreased in prominence in recent decades (e.g., textile industries), have 'too many' establishments (and a low average per establishment), or have 'too few' establishments to permit tabulation or meaningful analysis (e.g., potash, soda, and borate mining). To this list, two important industries were added that were otherwise nonexistent in 1983: ethyl alcohol manufacturing (NAICS 325193) and bottled water manufacturing (NAICS 312112).

Table 5 shows the 53 NAICS industries that are in scope to the water use inquiry in the 2017 EC. There are 41 industries in manufacturing and 12 in mining.[36] According to the 2013 County Business Patterns, these industries collectively had 39,121 establishments (27,901 in

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time of the final SWUM surveys, there were roughly 10,000 manufacturing establishments and 1,500 mining facilities with at least 20 million gallons of water intake (Becker 2016).

[35] The 1983 water intake data found in U.S. Bureau of the Census (1985, 1986) is suppressed for a non-trivial number of industries, for disclosure (or complimentary disclosure) reasons or because they had trivial water use. Collectively, these "missing" industries accounted for about 4.5% of water intake (in manufacturing and mining). Unfortunately, there may be some important water-using industries in this group. In the end, there are 277 manufacturing and mining industries – i.e., four-digit Standard Industrial Classification (SIC) codes – with non-missing, non-suppressed data on 1983 water intake. For tractability, we focused on just the top 62 industries, which accounted for 90% of total water intake.

These data were then converted to the most current industrial classification. Specifically, these data were converted from the 1972 SIC basis to the 1987 SIC basis, and then on to the 1997 North American Industrial Classification System (NAICS), and finally to the 2012 NAICS basis, taking care of the one-to-many, many-to-one, and many-to-many splits and joins along the way and applying the appropriate weights for all these splits and joins.

This NAICS-based listing of large water-using industries was then reviewed by the hydrologists from the USGS' National Water Use Leadership Team in developing a prioritization of industries. They considered whether some industries should perhaps rank higher than their outdated numbers suggest, perhaps because the industry has risen in importance or for geographic or other considerations. Likewise, they considered whether some industries should perhaps rank lower, perhaps because the industry has declined in importance and/or it has a relatively low average intake per establishment (but a large number of establishments).

[36] Because all three coal mining industries (in NAICS 21211) are covered by a single survey instrument (#21210), anthracite mining (NAICS 212113) will also be included here. Meanwhile, there are two survey instruments (#21301 and #2130A) for drilling oil and gas wells (NAICS 213111), to accommodate "alternative reporting units" in this industry. Otherwise, for all other cases, there is only one industry per survey instrument and one survey instrument per industry.

manufacturing, 11,220 in mining). If the 1983 statistics are a good approximation, these industries collectively account for about 90% of total water intake in these sectors.[37]

5.4 Post-Collection Plans: Industrial Water Use

CES staff, together with federal stakeholders and other interested parties, will validate and evaluate the quality of these new data and develop suitable edit and imputation methodologies for these items, with the goal of releasing certain experimental tables, where possible. For example, quantities that could be tabulated and published include: (i) gallons of water intake, total and by main source, and (ii) gallons of gross water use, total and by main source, where main source is one of seven possibilities: public water system – potable, public water system – reclaimed, self-supplied surface water, self-supplied ground water, self-supplied tide water, self-supplied mine water, and other. To the extent possible, these quantities could be tabulated by detailed industry, by state, and by industry group by state. Tabulations by other geographies (e.g., major watersheds) might also be desirable, as well as tabulations that add business size (e.g., employment) as a dimension. Tabulations might also include the share of businesses that recirculate/reuse water. (The quantity of recirculate/reused water is simply the difference between the two quantities above: gross water use minus water intake.) The results of these efforts will be made public through a CES working paper. Important contextual data on total employment, output, and other variables, tabulated by detailed industry, state, and so forth, will be published in due course as part of usual EC operations.

In addition, we hope that one or more research projects that use the confidential, establishment-level microdata will be initiated, by CES economists and researchers using the FSRDCs, to take a deeper look at water use and conservation in U.S. manufacturing and mining. This research may make use of other business microdata collected by the Census Bureau, as well as relevant facility-level data that may exist outside the Census Bureau, such as at state agencies.

6. Concluding Remarks

Through its publications, research programs, and outreach to stakeholders, the Census Bureau strives to continually improve its measures of the economy. This paper describes four lines of inquiry added to the recent Economic Census in an attempt to fill important data gaps through the efforts of researchers at the Center for Economic Studies. CES economists undertake research using confidential Census Bureau business microdata in order to increase our understanding of the economy and to provide insights into the quality of the data produced by the Census Bureau, sometimes leading to suggestions for new content on existing surveys, enhancements to survey methodology, and occasionally entirely new survey instruments.

[37] The top five industries alone account for perhaps 53%, the top 15 for about 75%, and the top 33 (those with over 45 billion gallons) for about 85%.

As of the writing of this paper, the important steps of question development, testing, and collection are complete. The next important steps are validating the data and determining optimal and feasible dissemination strategies.

We plan to issue a CES Working Paper for each area of inquiry, describing post-collection empirical work that aims to assess the quality of the data collected. In some cases, data quality will be sufficiently high, and disclosure concerns sufficiently minor, to allow us to possibly publish experimental data tabulations. When it is not possible to do so, the working paper will outline the challenges that precluded experimental tables. Finally, we expect all EC microdata will be made available to qualified researchers on approve projects in the FSRDC network so that they too may be able to add to our stock of knowledge about these new lines of inquiry.

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Table 1. Retail Health Clinics in the 2017 EC: Questions, Instructions, and NAICS Industries in Scope

445110 Supermarkets and Other Grocery (except Convenience) Stores
446110 Pharmacies and Drug Stores
452210 Department Stores
452311 Warehouse Clubs and Supercenters

1. Did this establishment operate or lease space to a retail health clinic in 2017?

   A retail health clinic is an in-store clinic with a health care professional who provides medical care (e.g., vaccines, health screenings, treatment of minor injuries and illnesses, or management of medications and treatments).

   □ Yes
   □ No

2. [If YES in question 1] Which ONE of the following best describes who operated this retail health clinic in 2017?

   □ Operate own retail health clinic
   □ Lease space to other business or health system that operates the retail health clinic
   □ Partner or jointly operate the retail health clinic with another business or health system

3. [If OPERATE or PARTNER in question 2] What was the total revenue from patient care services in 2017? *(Report this figure net of any negotiated discounts and write-downs for bad debt.)*

   $ _ _ _ , _ _ _ ,000.00

   Include:
   - *The value of total patient care operating receipts collected for providing medical services, such as flu shots, immunizations, diagnosis and treatment of sore throats, and chronic disease screening*
   - *Revenues from government payers (Medicare, Medicaid, private insurance – group, employer-sponsored, and all out-of-pocket costs including deductibles and co-insurance from private and public (Medicare/Medicaid) paid by the beneficiary or the family of the beneficiary)*

   Exclude:
   - *Non-patient care revenues from items such as food, clothing, and entertainment*
   - *Revenues from the sale of durable medical equipment, prosthetics, and orthotics*

4. [If OPERATE or PARTNER in question 2] Which of the following patient-care services did this retail health clinic offer in 2017? *Select ALL that apply.*

   □ Flu shots
   □ Immunizations other than flu shots, such as travel immunizations
   □ Other preventive health care services, including biometric screenings or lab tests

□ Mild acute care, for example diagnosis and treatment of sore throats or minor skin conditions
□ Chronic disease screening, monitoring, and/or management for conditions such as hypertension, diabetes, high cholesterol, or asthma
□ Behavioral health screenings to help identify common mental health conditions
□ Weight management programs
□ Pharmacotherapy management program, which may include delivery of medications and consultation to hospital patients
□ Information on health insurance options
□ Other patient services – Describe {write-in box}

5. [If OPERATE or PARTNER in question 2] Which of the following were used by the retail health clinic in 2017? *Select ALL that apply.*

□ Electronic health records
An electronic health record is an electronic version of a patient's medical history and may include their date of birth/gender, medical history, medications, immunizations, etc.

□ ePrescribing
With ePrescribing or electronic prescribing, a physician enters information about drugs a patient needs into a computer, and sends this electronic prescription to the patient's pharmacy for filling.

□ Telemedicine
Telemedicine refers to the remote diagnosis, monitoring, and/or treatment of patients by means of telecommunications technology.

□ Mobile coaching apps
Health care providers use mobile coaching apps to help remotely track their patients' compliance and progress with treatment.

□ Interactive patient-interview software
This software allows patients to use questionnaires to provide their medical information electronically to the health care provider prior to a visit.

6211 Offices of Physicians
6213 Offices of Other Health Practitioners
6214 Outpatient Care Centers

1. Was this establishment involved in the operation of a retail health clinic in 2017?

□ Yes
□ No

Table 2. Management Practices in Health Care Services in the 2017 EC: Questions, Instructions, and NAICS Industries in Scope

6211 Offices of Physicians
6212 Offices of Dentists
6213 Offices of Other Health Practitioners
6214 Outpatient Care Centers
6215 Medical and Diagnostic Laboratories
6216 Home Health Care Services
6219 Other Ambulatory Health Care Services
6221 General Medical and Surgical Hospitals
6222 Psychiatric and Substance Abuse Hospitals
6223 Specialty Hospitals
6231 Nursing Care Facilities
6232 Residential Mental Retardation, Mental Health and Substance Abuse Facilities
6233 Community Care Facilities for the Elderly
6239 Other Residential Care Facilities

1. Who sees your organization's measures of clinical performance? *Select ALL that apply.*

   Measures of clinical performance include counts, incidence rates, and other measures of specific clinical processes and outcomes.

   □ Managers
   □ Employees (non-managers)
   □ Patients and their responsible parties
   □ On public display

2. Who chooses which measures of clinical performance to collect? *Select ALL that apply.*

   □ Managers at this establishment
   □ Managers at other establishments and/or headquarters
   □ Insurance providers
   □ Government regulators or agencies
   □ Board of Directors

3. How frequently did senior management at this organization review the measures of clinical performance?

   □ Yearly or quarterly
   □ Monthly or weekly
   □ Daily or more often
   □ Never

Table 3. Self-service and Extra Services in Retail and Services in the 2017 EC: Questions and NAICS Industries in Scope

444110 Home Centers
445110 Supermarkets and Other Grocery (except Convenience) Stores
445120 Convenience Stores
446110 Pharmacies and Drug Stores
452210 Department Stores
452311 Warehouse Clubs and Supercenters
452319 All Other General Merchandise Stores

1. As of December 31, 2017, did this establishment offer a dedicated self-checkout lane for customers?
   □ Yes
   □ No

722511 Full-Service Restaurants
722513 Limited-Service Restaurants
722514 Cafeterias, Grill Buffets, and Buffets
722515 Snack and Nonalcoholic Beverage Bars

1. As of December 31, 2017, did this establishment use electronic devices for self-service table orders and/or payment (for example, through the use of a restaurant provided electronic device at the table)?
   □ Yes
   □ No

445110 Supermarkets and Other Grocery (except Convenience) Stores

1. As of December 31, 2017, did this establishment offer preordering or delivery services by website, app, fax, phone, or other means?
   □ Yes
   □ No

Table 4. **Water Use in Manufacturing and Mining in the 2017 EC: Questions and Instructions**

1. What was this establishment's water intake in 2017, that is, the quantity of new water introduced into the establishment for the first time, regardless of source or quality?

   Include water used in the production process and auxiliary operations (such as cooling and condensing, boiler feed, sanitary and domestic use). Report to the nearest thousand.

   _ , _ _ _ , _ _ _ , _ _ _ ,000 gallons

2. Did this establishment recirculate or reuse any water during 2017?

   □ Yes
   □ No

3. [If YES in question 2] What was this establishment's gross water use in 2017, that is, the quantity of water that would have been required if no water had been recirculated or reused?

   For example, if total water intake was 500 million gallons and, of these 500 million gallons, 100 million gallons were used twice for cooling purposes and once for washing products or materials, the total water required would be 300 million gallons, plus the 400 million gallons not recirculated, for a total of 700 million gallons of gross water use (less consumption and evaporation loss). Report to the nearest thousand.

   _ , _ _ _ , _ _ _ , _ _ _ ,000 gallons

4. What was this establishment's main source of new water in 2017? *Select only ONE.*

   <u>Public water system</u> includes water supplied by a water utility (whether municipally- or privately-owned) whose primary purpose is the supply of water to the general public and/or industrial users. <u>Self-supplied water systems</u> include water obtained by this establishment through its own system of pumps, pipes, hoses, etc. Also include here water obtained from another company that is not primarily a water utility.

   □ Public water system (municipally- or privately-owned utility)
   □ Self-supplied surface water system (rivers, streams, lakes)
   □ Self-supplied ground water system (wells, deep springs)
   □ Self-supplied tide water system (estuaries, bays, oceans)
   □ Self-supplied mine water (underground mines, quarries, open pits, water produced with oil)
   □ Other sources (e.g., rainwater, truck deliveries)

5. [If PUBLIC WATER SYSTEM in question 4] Which of the following types of water was mainly supplied by the public water system? *Select only ONE.*

   □ Potable
   □ Reclaimed wastewater

Table 5. Water Use in Manufacturing and Mining in the 2017 EC: NAICS Industries in Scope

Mining	211111	Crude Petroleum and Natural Gas Extraction
	211112	Natural Gas Liquid Extraction
	212111	Bituminous Coal and Lignate Surface Mining
	212112	Bituminous Coal Underground Mining
	212113	Anthracite Mining
	212210	Iron Ore Mining
	212234	Copper Ore and Nickel Ore Mining
	212312	Crushed and Broken Limestone Mining and Quarrying
	212321	Construction Sand and Gravel Mining
	212322	Industrial Sand Mining
	212392	Phosphate Rock Mining
	213111	Drilling Oil and Gas Wells
Manufacturing	311221	Wet Corn Milling
	311224	Soybean and Other Oilseed Processing
	311314	Cane Sugar Manufacturing
	311411	Frozen Fruit, Juice, and Vegetable Manufacturing
	311421	Fruit and Vegetable Canning
	311422	Specialty Canning
	311511	Fluid Milk Manufacturing
	311611	Animal (except Poultry) Slaughtering
	311615	Poultry Processing
	312112	Bottled Water Manufacturing
	312120	Breweries
	321113	Sawmills
	321912	Cut Stock, Resawing Lumber, and Planing
	322110	Pulp Mills
	322121	Paper (except Newsprint) Mills
	322122	Newsprint Mills
	322130	Paperboard Mills
	322299	All Other Converted Paper Product Manufacturing
	324110	Petroleum Refineries
	325110	Petrochemical Manufacturing
	325120	Industrial Gas Manufacturing
	325130	Synthetic Dye and Pigment Manufacturing
	325180	Other Basic Inorganic Chemical Manufacturing
	325193	Ethyl Alcohol Manufacturing
	325194	Cyclic Crude, Intermediate, and Gum and Wood Chemical Manufacturing
	325199	All Other Basic Organic Chemical Manufacturing
	325211	Plastics Material and Resin Manufacturing
	325212	Synthetic Rubber Manufacturing
	325220	Artificial and Synthetic Fibers and Filaments Manufacturing
	325311	Nitrogenous Fertilizer Manufacturing
	325312	Phosphatic Fertilizer Manufacturing
	325320	Pesticide and Other Agricultural Chemical Manufacturing
	325411	Medicinal and Botanical Manufacturing
	325998	All Other Miscellaneous Chemical Product and Preparation Manufacturing
	326199	All Other Plastics Product Manufacturing
	327310	Cement Manufacturing
	331110	Iron and Steel Mills and Ferroalloy Manufacturing
	331210	Iron and Steel Pipe and Tube Manufacturing from Purchased Steel
	331313	Alumina Refining and Primary Aluminum Production
	331315	Aluminum Sheet, Plate, and Foil Manufacturing
	331511	Iron Foundries